Thien Nguyen

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SUMMARY

Engineer with 5 years of systems integration experience and a strong foundation in fluid, thermal, and electro-mechanical modeling. Design and validate FEA, CFD, and analytical models, automate analysis in Python, and drive results to decisions.

CERTIFICATIONS

Certified SolidWorks Professional (**CSWP**), Project Management Professional (**PMP**), Geometric Dimensioning & Tolerancing Fundamentals (**GD&T Fundamentals**)

TECHNICAL SKILLS

Analysis & Modeling: SolidWorks Flow Simulation (CFD), Finite Element Analysis (FEA), analytical methods, thermal & fluid analysis, mechanical testing & validation

Programming: Python, SQL, C, C++

Tools & Software: SolidWorks (CSWP), MS Office Suite (Excel, Word, PowerPoint), 3D printing, embedded systems, KiCad

Relevant Coursework: Thermodynamics, Fluid Dynamics, Materials Science, Data Structures, Computational Algorithms

EXPERIENCE

Technical Project Manager — *August 2021 - Present*

DH Pace Company Inc. — *Phoenix, AZ*

National security systems integration company with 50+ U.S. locations, serving commercial, industrial, and institutional sectors.

Engineering

- Developed a Python tool to automate telemetry signal verification, replacing manual review to improve accuracy and reallocate engineering effort to higher-value work.
- Built an automated reminder system that flags license expirations and triggers proactive customer outreach, contributing to $1M+ in annual revenue retention across multiple offices.
- Commissioned and integrated systems end-to-end, resolving hardware and software incompatibilities across migrations and providing cross-platform technical support, with most issues resolved same-day.
- Validated access control wiring against datasheet schematics prior to field deployment, eliminating field errors and rework.

Project Management

- Led 20+ concurrent projects as the single point of contact across customers, onsite technicians, and third-party vendors, translating technical requirements between stakeholders and driving each project from proposal handoff through final city inspection under tight timelines and within budget.
- Wrote standard operating procedures for installation and commissioning workflows, adopted across multiple offices to standardize execution and reduce onboarding time for new technicians.

PROJECTS - khathien.github.io/projects

LFP Battery Thermal Analysis | SolidWorks Flow Simulation, CFD, Heat Transfer

- Conducted steady-state CFD analysis of a commercial LFP cell under natural convection to establish maximum temperature baselines, scaling from a single cell to a 100-cell stack within a properly sized computational domain.
- Documented assumptions, limitations, and next steps (forced convection, transient analysis), with conservative material properties and ASME heat-transfer literature cited throughout.

Pipe Flow, Heat & Mass Transfer | SolidWorks Flow Simulation, CFD, Fluid Dynamics, Analytical Methods

- Simulated laminar and turbulent velocity profiles in an intersecting pipe network, confirming laminar inflow against the Reynolds number threshold and validating CFD results against analytical theory curves.
- Modeled a double-pipe heat exchanger in co-current and counter-current orientations, calculating overall heat transfer coefficients analytically and comparing them against simulation to evaluate CFD accuracy.

Self-driving Trashcan Robot | Python, C++, SolidWorks, DFA, 3D Printing, Embedded Systems

- Developing the autonomous Phase 2 navigation in Python on a Raspberry Pi, building on the completed C++ firmware that runs PWM motor control and ESP-NOW wireless communication across ESP32 controllers.

EDUCATION

Grand Canyon University — *Phoenix, AZ | April 2021*

Bachelor of Science, Mechanical Engineering, GPA: 3.99

Arizona State University — *Tempe, AZ | In Progress*

Bachelor of Science, Computer Science